

Erdős Problem #462

EP-462

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $p(n)$ denote the least prime factor of n . There is a constant $c > 0$ such that $\sum_{\substack{n < x \\ n \text{ not prime}}} \frac{p(n)}{n} \sim c \frac{x^{1/2}}{(\log x)^2}$.

Is it true that there exists a constant $C > 0$ such that $\sum_{x \leq n \leq x + Cx^{1/2}(\log x)^2} \frac{p(n)}{n} \gg 1$ for all large x ?

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”difficulty”: ”L1”

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>